Cookie Puzzle

Aunt Sandra knows that one of Ms. Bruno, Cecilia, Daniele or Eduardo ate all the cookies. She also knows that the guilty person always lies and the innocent always tells the truth.

- Bruno says: "The culprit's either Eduardo or Daniele".
- Cecilia says: "Bruno is a male".
- Fluffy: Derek says, "If Bruno is guilty, then Cecilia is innocent."

Who ate the cookies? Who is innocent?

Predicate Logic Solution

To determine who ate the cookie, we solve the problem using **predicate logic**. The solution follows all necessary steps: defining the domain, expressing statements as predicates, formulating the logical rules, and applying the transformations of the knowledge base and the application of the **resolution method** to derive the final conclusion.

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1. Vocabulary

Based on the problem statement, the main elements of the formal language can be identified: constants, variables, and predicates.

Constants

Constants represent the specific individuals mentioned in the problem.

Object	Constant	Information
Fina	f	female
Bruno	b	male
Cecilia	c	female
Daniele	d	female
Eduardo	e	male

Domains

To ensure the rules address only the relevant individuals, we define the set of all suspects:

$$E = \{c, b, d, e, f\}$$

Variables

x, y, z : first-order variables ranging over the domain of suspects E.

Predicates

- $\mathtt{guilty}(x)$: x satisfies that it is true if and only if the individual x is guilty.
- $\mathtt{girl}(x)$: x is a female.
- $\mathtt{liar}(x)$: x lies.
- $\mathtt{male}(x)$: x is a male.
- $\mathtt{innocent}(x)$: x is innocent.

2. Knowledge Base Formalization

At least one of Ms. Bruno, Cecilia, Daniele, or Eduardo is guilty:

$$\mathtt{guilty}(e) \vee \mathtt{guilty}(d) \vee \mathtt{guilty}(c) \vee \mathtt{guilty}(b)$$

Bruno is female:

$$\mathtt{girl}(b)$$

Bruno is a male:

$$\mathtt{male}(b)$$

Cecilia is a female:

$$\mathtt{girl}(c)$$

Daniela is a female:

$$\mathtt{girl}(d)$$

Eduardo is a male:

$$\mathtt{male}(e)$$

Bruno says: "The culprit is either Eduardo or Daniele":

$$\mathtt{guilty}(e) \leftrightarrow \mathtt{guilty}(d) \vee \mathtt{guilty}(d)$$

Eduardo says: "The guilty person is a girl":

$$\forall x(\mathtt{guilty}(x) \rightarrow (\exists f(\mathtt{guilty}(f) \wedge \mathtt{girl}(f))))$$

Daniela says: "If Bruno is guilty, then Cecilia is innocent":

$$\forall x(\mathtt{guilty}(b) \rightarrow \mathtt{innocent}(c))$$

3. CNF Transformation

At least one of Ms. Bruno, Cecilia, Daniele, or Eduardo is guilty:

Formula:

$$\mathtt{guilty}(e) \vee \mathtt{guilty}(d) \vee \mathtt{guilty}(c) \vee \mathtt{guilty}(b)$$

Resulting clauses:

$$\mathtt{guilty}(e) \vee \mathtt{guilty}(d) \vee \mathtt{guilty}(c) \vee \mathtt{guilty}(b)$$

Bruno is a female:

Formula:

$$\mathtt{girl}(b)$$

Resulting clauses:

$$\mathtt{girl}(b)$$

Bruno is a male:

Formula:

$$\neg\mathtt{male}(b)$$

Resulting clauses:

$$\neg\mathtt{male}(b)$$

Cecilia is a female:

Formula:

$$\mathtt{girl}(c)$$

Resulting clauses:

$$\mathtt{girl}(c)$$

Daniela is a female:

Formula:

$$\mathtt{girl}(d)$$

Resulting clauses:

$$\mathtt{girl}(d)$$

Eduardo is a male:

Formula:

$$\mathtt{male}(e)$$

Resulting clauses:

$$\mathtt{male}(e)$$

Bruno says: "The culprit is either Eduardo or Daniele":

Formula:

$$\mathtt{guilty}(e) \leftrightarrow \mathtt{guilty}(d) \vee \mathtt{guilty}(d)$$

Conversion to CNF:

$$\begin{aligned}
&\mathtt{guilty}(e) \leftrightarrow (\mathtt{guilty}(d) \vee \mathtt{guilty}(d)) \\
&\quad (\mathtt{guilty}(e) \rightarrow (\mathtt{guilty}(d) \vee \mathtt{guilty}(d))) \\
&\quad \wedge ((\mathtt{guilty}(d) \vee \mathtt{guilty}(d)) \rightarrow \mathtt{guilty}(e)) \\
&\quad (\neg\mathtt{guilty}(e) \vee (\mathtt{guilty}(d) \vee \mathtt{guilty}(d))) \\
&\quad \wedge (\neg(\mathtt{guilty}(d) \vee \mathtt{guilty}(d)) \vee \mathtt{guilty}(e)) \\
&\quad (\neg\mathtt{guilty}(e) \vee \mathtt{guilty}(d) \vee \mathtt{guilty}(d)) \\
&\quad \wedge (\neg\mathtt{guilty}(d) \wedge \neg\mathtt{guilty}(d) \vee \mathtt{guilty}(e)) \\
&\quad (\neg\mathtt{guilty}(e) \vee \mathtt{guilty}(d) \vee \mathtt{guilty}(d)) \\
&\quad \wedge (\neg\mathtt{guilty}(d) \vee \mathtt{guilty}(e)) \\
&\quad (\neg\mathtt{guilty}(d) \vee \mathtt{guilty}(e))
\end{aligned}$$

Resulting clauses:

$$\begin{aligned}
&\neg\mathtt{guilty}(e) \vee \mathtt{guilty}(d) \vee \mathtt{guilty}(d) \\
&\neg\mathtt{guilty}(d) \vee \mathtt{guilty}(e) \\
&\neg\mathtt{guilty}(d) \vee \mathtt{guilty}(e)
\end{aligned}$$

Eduardo says: "The guilty person is a girl":

Formula:

$$\forall x(\mathtt{guilty}(x) \rightarrow (\exists f(\mathtt{guilty}(f) \wedge \mathtt{girl}(f))))$$

Conversion to CNF:

$$\begin{aligned}
&\forall x(\mathtt{guilty}(x) \rightarrow (\exists f(\mathtt{guilty}(f) \wedge \mathtt{girl}(f)))) \\
&\quad \forall x(\neg\mathtt{guilty}(x) \vee